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ControlScene: Controllable Text-to-3D Scene Generation via Structured Layout Priors

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ABSTRACT

Text-to-3D scene generation has emerged as a promising approach for enabling natural and intuitive 3D content creation through language. However, existing methods often struggle with generating scenes that are semantically faithful, spatially coherent, and controllable, primarily due to the lack of grounded spatial reasoning and fine-grained structural supervision. In this paper, we present ControlScene: Controllable Text-to-3D Scene Generation with Layout Guidance, a novel framework that incorporates layout-level structure into the generation process, allowing LLM to jointly reason about semantics and spatial composition. To support this framework, we introduce LayoutVerse-20K, a large-scale benchmark dataset comprising 20,000 manually annotated samples, each including textual prompts, layout metadata, graph-structured layouts, scene graphs, and corresponding 3D scenes. For fine-grained evaluation, we propose two task-specific metrics: category and layout plausibility, which assess semantic and spatial alignment by comparing generated outputs against commonsense priors derived from Top-K reference samples. Additionally, we develop an interactive user interface for real-time layout customization, enhancing controllability and practical usability. Extensive experiments show that ControlScene significantly outperforms baseline methods in spatial realism, semantic consistency, and user controllability, providing a robust foundation for future research in grounded, language-driven 3D scene generation. All experimental codes of this work are open-source at: Project site: <https://winkquan.github.io/ControlScene-Page>.

1 | Introduction

Generating 3D scenes from natural language represents a significant advancement toward accessible 3D content creation. This capability has broad applications in virtual reality, gaming, interior design, and simulation, where flexible and efficient scene design is crucial. Text-driven generation provides a powerful interface enabling users, regardless of technical expertise, to create spatially rich environments using high-level linguistic input. Recent advances in large language model (LLM) and generative modeling have renewed interest in this field. Methods such as DreamFusion [1] and Text2Room [2] demonstrate the feasibility of generating 3D content from textual prompts, primarily focusing on object-level reconstruction or room-scale synthesis using

neural radiance fields. Other approaches like 3D-SceneGen [3] and SceneScript [4] have attempted to structure textual input into scene graphs or symbolic layout plans to support more controllable generation. These structures facilitate more interpretable scene generation.

Recent advancements in LLM [5] and generative techniques [6, 7] have enabled the generation of semantically rich 3D content from textual descriptions. However, the field still faces significant challenges in producing realistic and contextually coherent scenes. A primary challenge is the Penrose Triangle Problem, which involves generating outputs that, while syntactically correct, contradict common sense or real-world logic. For example, a prompt like "a cat sitting on a cloud inside a bedroom" might be taken literally, resulting in a scene that defies physical plausibility. These

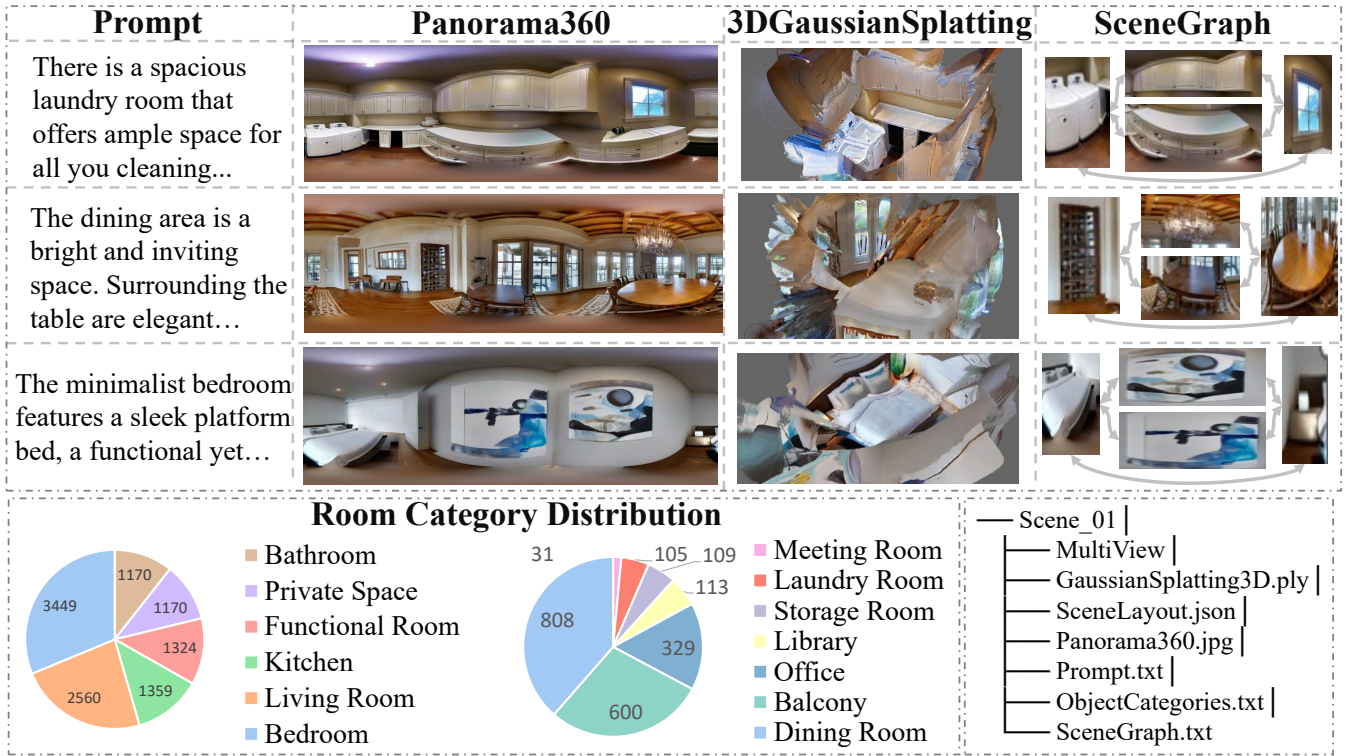


FIGURE 1 | Overview of the proposed LayoutVerse-20K dataset. Each sample includes a natural language prompt, a corresponding panoramic picture, a 3D scene representation, and a structured SceneGraph capturing spatial and semantic relationships. The bottom section shows the distribution of room categories and the organized dataset file structure.

kinds of results highlight a disconnect between fluent language generation and grounded spatial reasoning. Moreover, LLM lack an innate understanding of visual attributes such as color, shape, proportion, and spatial relationships. This shortcoming can lead to inconsistencies in generated scenes, particularly when capturing intricate physical arrangements or maintaining stylistic coherence. Unlike humans, LLM do not visually perceive their outputs. They only rely on inferred patterns from data, often resulting in discrepancies between the intended descriptions and the actual visual representations [8].

Another critical challenge in text-driven 3D scene generation is the absence of standardized, richly annotated datasets that are tailored to capture the intricate correspondence between language and spatial understanding. Unlike traditional datasets with object-level annotations or simple scene descriptions, advancing this field requires multi-modal datasets incorporating layout metadata, scene graphs, and structured relational mappings among entities. These datasets should encode object presence and spatial configurations, relative positions, and functional relationships, such as "the bed is aligned parallel to the wall" or "the chair is tucked under the table." Such data is crucial for teaching models not just to populate scenes, but to compose them in ways that align with human intuition and spatial reasoning. Equally important is the lack of robust evaluation metrics for assessing the quality and accuracy of generated scenes. Existing benchmarks often emphasize visual fidelity or basic object inclusion, overlooking the deeper question: does the generated scene genuinely reflect the user's textual intent? To address this, we need alignment metrics that quantitatively assess the consistency between scene layouts and textual prompts [9]. These

could include layout-text alignment scores, graph similarity metrics comparing predicted scene graphs with ground truth, and semantic spatial correctness measures that evaluate whether objects are positioned in functionally and contextually appropriate ways.

To address the aforementioned challenges, we propose a novel framework ControlScene: Controllable Text-to-3D Scene Generation with Layout Guidance. Our approach integrates the contextual reasoning capabilities of LLM with structured spatial representations by embedding LLM directly into the scene layout generation pipeline. This enables the model to move beyond surface-level language understanding and engage in guided spatial composition grounded in visual semantics. To address LLM's inherent limitations in visual perception, especially regarding color, shape, scale, and spatial arrangements, we construct a large-scale benchmark dataset, LayoutVerse-20K, comprising 20,000 manually annotated samples. Each sample includes a textual prompt, detailed layout metadata, a layout graph, a scene graph, and the final 3D scene. As shown in Fig. 1, this dataset supports both effective training and benchmarking of spatial and semantic fidelity in generated scenes. To further assess the alignment of generated scenes with commonsense knowledge and textual intent [10], we introduce two novel metrics: category plausibility and layout plausibility. Category plausibility measures whether the object categories in a scene are semantically appropriate, based on priors derived from the Top-K most relevant scenes in the dataset. Layout plausibility evaluates whether the spatial arrangement of objects follows common, logical patterns by measuring structural similarity between the generated layout

graph and reference graphs. These metrics introduce greater interpretability and precision, surpassing traditional pixel-level or object-count evaluations. Finally, we develop an interactive user interface enabling real-time customization of scene layouts. This feature significantly enhances system flexibility, enabling users to guide the generation process according to specific needs or preferences, thus bridging automated generation with human creativity and intent. The main contributions can be summarized as follows:

- We propose a novel ControlScene framework that incorporates layout guidance into the generation process. By integrating LLM with structured spatial representations, our method enables controllable and semantically grounded 3D scene generation.
- We introduce two evaluation criteria: category plausibility and layout plausibility, which assess the semantic and spatial consistency of generated scenes by comparing them with statistically grounded priors derived from Top-K retrieved samples.
- We construct LayoutVerse-20K, a high-quality benchmark dataset containing 20,000 manually annotated samples. Each sample includes a textual prompt, layout metadata, layout graph structure, scene graph, and the corresponding 3D scene. This dataset offers a comprehensive resource for training, benchmarking, and advancing grounded 3D scene generation research.

2 | Related Work

Layout generation is a key task in automatic graphic design, covering areas such as interior design [11, 12] and graphical user interfaces [13]. Previous methods are mostly based on deep learning models, which need to be trained from scratch and are time-consuming. For example, LayoutGAN [14] generates wireframes of fixed elements, LayoutVAE [15] generates layouts based on object labels, and LayoutTransformer [16] uses self-attention mechanisms to complete layouts. However, these methods are limited to class labels and cannot handle numerical and spatial concepts under text conditions. With the development of large model technology, large language models have shown great potential in layout generation. LayoutGPT [17] uses large language model technology to efficiently convert text into two-dimensional layouts, significantly reducing cost and time and improving generation efficiency. Recent studies have also confirmed the effectiveness of such methods. Inspired by these works, this study proposes a layout generation method based on large language models that can quickly generate high-quality layouts without additional training, improving efficiency and flexibility.

Room scene generation requires the creation of 3D content and its reasonable placement. Traditional methods usually first construct a set of 3D objects and then optimize the layout of these objects through iterative methods [18, 19], nonlinear optimization [20], or manual interaction [21]. However, these methods are often time-consuming and rely on manual intervention, making it difficult to achieve efficient and diverse scene generation. With the development of large-scale indoor scene datasets [22], recent research has turned to using generative networks to automate this process. These methods use advanced techniques such

as feedforward networks [23], variational autoencoders (VAEs) [24], generative adversarial networks (GANs) [25], autoregressive models [26], and diffusion models [27] to generate diverse and realistic 3D scenes. However, these methods usually have difficulty handling objects that do not appear in the training dataset, resulting in limitations in the diversity and adaptability of the generated scenes. At the same time, researchers in recent years have found that the generated scenes seem reasonable in terms of local 3D structure, but have defects in global consistency. This type of problem is called the “Penrose Triangle Problem”. Recent studies [28] have attempted to overcome these limitations by introducing LLM. These studies use LLM to generate a style sheet language for object placement, providing flexible layout guidance for scene generation. Our work builds on this by generating a preliminary layout using LLM and further refining the scene using a diffusion model. We leverage the semantic understanding and detail generation capabilities of the model to generate high-quality 3D scenes and address the global consistency issue.

Language input has always been a core component of many vision-language tasks [29, 30]. As contemporary LLM have shown strong generalization capabilities, recent research works have begun to explore their potential in multimodal tasks [31, 32]. For example, VisProg [33] and ViperGPT [34] use LLM to design modular pseudocode instructions or executable Python programs to achieve visual reasoning tasks; LLMscore [35] uses LLM to evaluate the performance of text-to-image models; and Visual ChatGPT [36] proposes a prompt manager to support the execution of multiple image generation models. Although LLM have unique advantages in generation tasks, their lack of understanding of visual elements and spatial concepts may affect the quality of generation results. To this end, we propose a new method to integrate LLM into visual layout design and manually annotate 20,000 data containing text prompts, layout information, and scene graphs to make up for its shortcomings. Based on these data, we propose two new metrics to more accurately measure the generation quality and develop an interactive interface that allows users to customize the layout. These improvements are designed to improve LLM’s performance in visual layout generation, enhancing its understanding of visual elements and spatial relationships, while providing users with more freedom and flexibility.

3 | Method

As illustrated in Fig. 2, ControlScene adopts a two-stage pipeline to achieve controllable and semantically grounded scene generation. In the first stage, a LLM-driven layout generator empowers users to craft and iteratively refine scene layouts through an intuitive interactive interface. To overcome the inherent spatial reasoning limitations of LLM, we introduce LayoutVerse-20K, a purpose-built benchmark dataset used for spatial plausibility evaluation and model calibration. In the second stage, a diffusion-based model synthesizes a globally consistent panoramic image [37]. This is followed by a multi-view rendering pipeline involving super-resolution enhancement and spherical-to-perspective (e2p) projection. Finally, a detailed 3D reconstruction [38] is performed using 3D Gaussian Splatting, enabling faithful transformation from text prompts to high-fidelity, immersive 3D scenes.

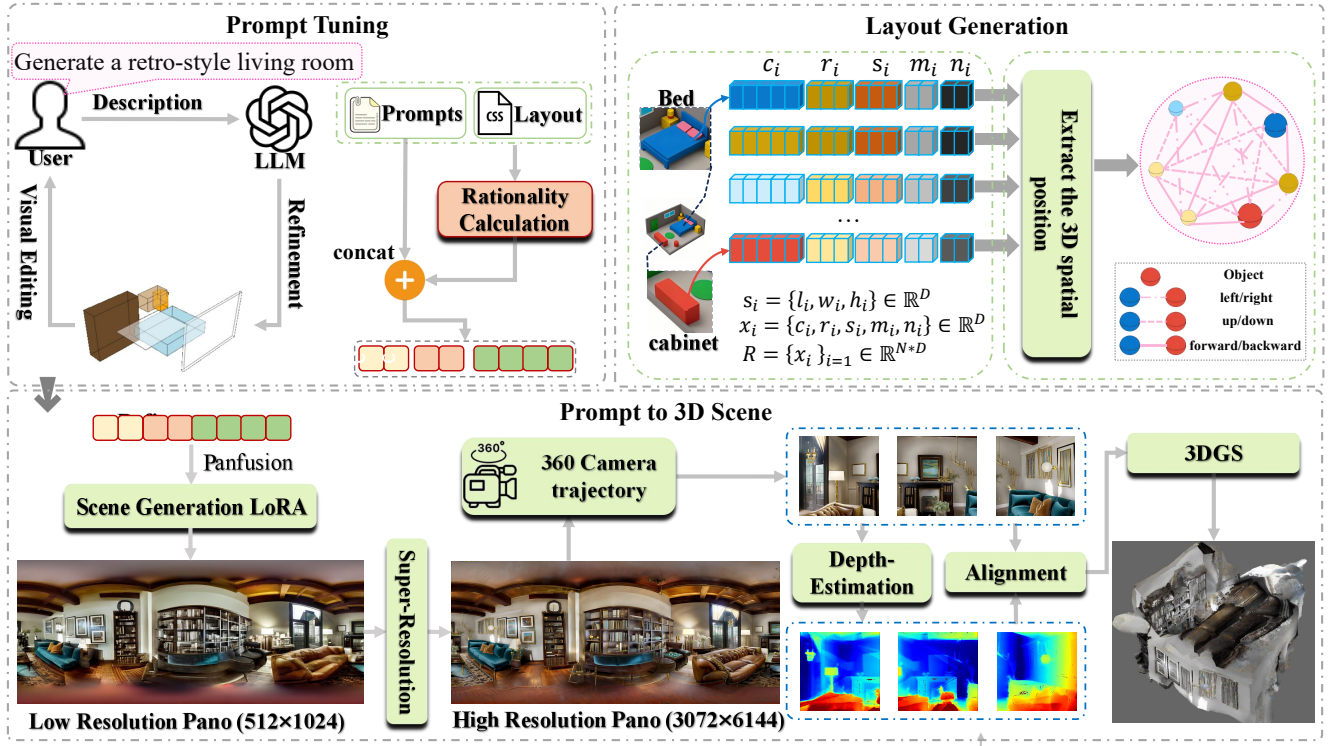


FIGURE 2 | Overview of the proposed ControlScene framework. The system takes a user prompt as input and performs layout generation via a LLM, followed by spatial rationality evaluation and refinement. The structured layout is then used to synthesize a panorama via a diffusion-based generator (Panfusion), followed by super-resolution, depth estimation, and 3D alignment to produce a high-fidelity 3D scene.

3.1 | Layout Generation

In text-driven 3D scene generation tasks, directly mapping unstructured natural language to continuous 3D pixels or Gaussian primitive representations faces enormous cross-modal semantic gaps and the lack of spatial logic. Although diffusion models provide powerful priors for visual feature generation and 3D Gaussian Splatting achieves extremely fast and high-fidelity rendering, both technologies essentially belong to the "generation" and "presentation" levels, lacking the ability of deep cognitive understanding and logical planning for complex indoor spatial topological relationships.

In current research, mainstream controllable generation solutions mostly attempt to introduce image sketches or coarse geometric layouts as guiding signals. However, this guiding mechanism often has two limitations: first, it imposes extremely high requirements on the scale and alignment accuracy of datasets. Developers usually need to spend substantial costs to construct perfectly matched data pairs of text, high-precision layouts, and 3D models; otherwise, the model is highly prone to generating "spatial hallucinations". Second, pure geometric guidance lacks self-correction capabilities at the semantic level. When user instructions conflict with physical common sense, traditional methods struggle to make logically consistent adjustments during the generation process.

To address the above challenges, this paper proposes deploying a LLM as the "spatial planning brain" in the ControlScene framework. Through a decoupled paradigm of "semantic planning, structural layout, and downstream instantiation", it utilizes the rich built-in common sense knowledge base of LLMs to replace part of the hard dependency on ultra-high-performance datasets,

and reduces the complexity of generation tasks through structured agents. This chapter will elaborate on how to realize highly editable 3D indoor scene layout generation based on LLMs. First, it introduces the formal modeling of scene tasks; then, it elaborates on the modeling process of LayoutVerse-20K, a large-scale prior dataset constructed to make up for the spatial cognitive limitations of large models; next, it discusses how to strengthen the topological associations between objects through graph structure modeling; on this basis, it focuses on the self-correction mechanism designed in this framework, which endows the model with the ability to identify and repair physical conflicts; finally, the superiority of the proposed method in terms of the rationality, coherence, and editing efficiency of layout generation is verified through detailed experiments.

3.1.1 | CSS Scene Code

To enable large language models, which primarily process one-dimensional discrete text sequences, to understand and generate 3D spatial layouts, it is necessary to perform dimensionality reduction and discretized mapping on the continuous physical space. This paper proposes a control-oriented scene representation termed CSS Scene Code (hereinafter referred to as scene sequence), which transforms scene layout generation into a structured sequence prediction task.

Specifically, inspired by the design philosophy of Cascading Style Sheets (CSS) in web front-end development, this work innovatively decouples complex 3D scene layouts and converts them into structured CSS-like scene sequences. A 3D scene layout L containing N furniture objects is defined as an ordered set of 3D

bounding boxes $L = \{O_1, O_2, \dots, O_N\}$. Each furniture object O_i is parameterized as a multi-dimensional attribute tuple:

$$O_i = (c_i, \mathbf{p}_i, \mathbf{s}_i, \mathbf{r}_i, \mathbf{m}_i, \mathbf{n}_i) \quad (1)$$

where c_i denotes the semantic category of the object; $\mathbf{p}_i = (x_i, y_i, z_i)$ represents the absolute central coordinate in the 3D world coordinate system, forming the basis for spatial localization; $\mathbf{s}_i = (l_i, w_i, h_i)$ describes the physical boundary dimensions along three orthogonal axes, i.e., length, width and height, which determine the spatial occupancy volume; $\mathbf{r}_i = (\theta_i)$ denotes the rotation angle around the vertical axis to constrain object orientation and layout alignment; $\mathbf{m}_i$ represents material and appearance attributes such as texture type and color preference, providing explicit visual guidance for downstream high-fidelity rendering pipelines; $\mathbf{n}_i$ is the discrete semantic category label that defines the basic functional class of each object, e.g., sofa, coffee table.

Through such structured decomposition of multi-dimensional attributes, the continuous and complex 3D physical space is effectively reduced and mapped into a one-dimensional textual attribute sequence that is easy for large language models to understand, reason about, and generate. This not only endows the model with fine-grained editable capability for individual objects but also provides fundamental support for spatial logical consistency of the global scene. With this sequential reconstruction, the generation process from natural language instruction T to 3D layout L is strictly equivalent to the next-token autoregressive prediction task of large language models.

3.1.2 | Graph Structure Modeling

In real-world 3D indoor spaces, high-quality scene construction relies not only on the precise placement of individual physical entities but also heavily on rigorous geometric relationships and functional semantic constraints among furniture items. An indoor environment is by no means a random aggregation of isolated objects; rather, it constitutes an organic whole governed by intricate hierarchical dependencies (e.g., a table lamp must be supported by a bedside table) and spatial topological relationships (e.g., dining chairs should surround a table symmetrically and equidistantly). Nevertheless, Large Language Models (LLMs) inherently operate on a one-dimensional autoregressive decoding paradigm. When processing complex indoor scenes containing numerous objects, relying solely on one-dimensional coordinate attribute sequences forces the model to implicitly memorize and infer all relative spatial rules within an extensively long context. Such implicit reasoning is highly susceptible to "attention drift" during the later stages of generation, making it challenging to maintain global topological coherence. Consequently, the generated layouts often suffer from disordered, isolated furniture arrangements, and even severe violations of daily living common sense. To overcome the inherent limitations of one-dimensional sequential representations in capturing long-range spatial geometric relationships, this section introduces explicit graph structure modeling [39] into the layout generation framework. During the control instruction preprocessing stage, the framework first

leverages the semantic extraction capabilities of the LLM to convert the input text into a scene sequence, and subsequently constructs the corresponding graph structure matrix $G = (V, E)$ [40] via dedicated algorithms:

- **Node Set V :** Represents the physical entities in the scene, where each node $v_i \in V$ carries the category and attribute features of the object.
- **Edge Set E :** Represents the spatial topological and semantic relationships between objects. It denotes the spatial relationship between two connected nodes v_i and v_j .

The geometric relationship definition rules are detailed in Table 1.

In the extraction and modeling of topological relationships for 3D indoor scenes, the determination priority of vertical (Z-axis) relationships is assigned a significantly higher weight than that of horizontal (X- and Y-axes) relationships. This hierarchical prioritization is primarily dictated by the absolute constraints of the physical world and the inherent structural characteristics of indoor spaces: Firstly, Z-axis relationships are governed by the law of gravity—the most fundamental and unbreakable physical constraint. They determine the physical load-bearing states among entities and form the bottom-up, core hierarchical support skeleton of the entire scene graph. Secondly, the physical fault tolerance for vertical relationships is exceptionally low. While horizontal layouts exhibit substantial subjective flexibility and tolerance, any violation along the Z-axis leads to fatal physical common-sense errors, such as floating objects or severe interpenetration. Lastly, from the perspective of algorithmic efficiency, adopting a "Z-axis-first" dimensionality-reduction screening strategy leverages spatial elevation differences to achieve natural isolation among physical tiers. This directly filters out complex and redundant cross-tier horizontal collision detections. This hierarchical processing mechanism not only substantially reduces computational overhead but also rigorously ensures the accuracy and semantic rationality of the extracted 3D spatial topology from an underlying physical logic perspective.

In conclusion, the introduction and structural extraction of the 3D scene graph effectively bridge the inherent gap of 1D sequential generation in global topological expression, providing LLMs with explicit spatial guidance for handling complex hierarchical dependencies. At the physical execution level, this structured representation enables automated geometric algorithms to perform real-time detection and positional fine-tuning of spatial conflicts (e.g., interpenetration or invalid merging), ensuring physical self-consistency. At the deep logical level, the graph structure decouples the indoor scene into independent "Object" entities. This not only offers a concrete handle for fine-grained manipulation but also establishes a crucial bridge between 3D geometry and the LLM's built-in common-sense prior knowledge. Consequently, the system can conduct proactive "logical auditing" (e.g., identifying and rejecting absurd functional layouts like multiple beds in a study room), effectively curbing functional layout errors at the source and significantly enhancing scene rationality.

TABLE 1 | Physical Meanings of Relationship Values in the Graph Structure Matrix

Value	Reference Axis	Proximity State	Spatial Meaning (m relative to n)
0	—	Coincident	The object itself
1	Y-axis (Left/Right)	Separated	m is to the left of n with some distance
-1	Y-axis (Left/Right)	Separated	m is to the right of n with some distance
2	Y-axis (Left/Right)	In contact	m is to the left of n , its right surface touching n 's left surface
-2	Y-axis (Left/Right)	In contact	m is to the right of n , its left surface touching n 's right surface
3	Z-axis (Up/Down)	Separated	m is above n with some distance (floating)
-3	Z-axis (Up/Down)	Separated	m is below n with some distance
4	Z-axis (Up/Down)	In contact	m is above n , its bottom surface touching n 's top surface (supported)
-4	Z-axis (Up/Down)	In contact	m is below n , its top surface touching n 's bottom surface
5	X-axis (Front/Back)	Separated	m is in front of n with some distance
-5	X-axis (Front/Back)	Separated	m is behind n with some distance
6	X-axis (Front/Back)	In contact	m is in front of n , its back surface touching n 's front surface
-6	X-axis (Front/Back)	In contact	m is behind n , its front surface touching n 's back surface

3.2 | Dataset

Although Large Language Models (LLMs) have demonstrated exceptional proficiency in natural language processing, they encounter substantial bottlenecks in spatial generation tasks. Primarily trained on textual corpora, LLMs inherently lack the intuitive understanding and depth required to process spatial relationships and geometric configurations [41, 42], making it difficult to fully capture complex physical and visual semantics. Consequently, even when augmented with external knowledge bases, they are prone to generating inaccurate or physically implausible layouts. Furthermore, LLMs face inherent difficulties in interpreting multi-dimensional spatial cues—such as depth, distance, and perspective—which are essential for maintaining visual consistency and spatial rationality. Fundamentally, their inability to intuitively perceive real-world scenes or physically interact with the environment strictly limits their spatial reasoning capabilities and the overall fidelity of scene generation.

To address these intrinsic limitations and provide a robust foundation for spatial reasoning, we introduce LayoutVerse-20K, a large-scale, multi-dimensional cross-modal [43] benchmark dataset encompassing approximately 20,000 high-quality indoor layout samples. While detailed engineering specifications are extensively documented in the Appendix, its rigorous construction process is briefly outlined here. The scene sources are meticulously curated from public interior design platforms, premium assets within 3D modeling communities, and real-world panoramic images. To bridge the modality gap for LLMs, the annotation format is strictly standardized: each data pair uniformly includes natural language instructions, structured CSS Scene Codes, scene relation graphs, multi-view rendered images, and their corresponding 3D representations. Moreover, strict quality control is maintained through a "human-AI collaborative validation" mechanism. Automated scripts initially screen for physical penetrations and dimensional anomalies, followed by manual spot-checks to guarantee highly accurate alignment among textual semantics, local spatial parameters, and overall 3D rendering performance. Leveraging this comprehensive dataset, we propose two novel metrics—layout rationality and category rationality—to accurately quantify generation quality. Ultimately, we implement a

prompt-tuning-based self-correction mechanism, wherein evaluative feedback from these metrics is iteratively embedded to guide and optimize spatial adjustments during the generation process.

3.2.1 | LayoutVerse-20K

Previous datasets [44, 45, 46, 47, 48] have predominantly been constructed in the form of panoramic or RGB-D images, with only a limited subset providing detailed scene descriptions or fully realized 3D representations [49]. To address these limitations and explicitly clarify the construction process, the key attributes and advantages of LayoutVerse-20K can be detailed as follows:

Diverse Scene Sources. To construct a robust and high-quality benchmark [50], we collected raw data from diverse sources, including public interior design platforms, 3D modeling communities, and real-world panoramic imagery. Consequently, LayoutVerse-20K contains hundreds of common scene types, ranging from typical domestic spaces (e.g., living rooms and bedrooms) to functional and private areas (e.g., gyms, wine cellars, and e-sports rooms). The dataset encapsulates various design styles and cultural concepts—such as Bohemian, minimalist, and traditional Chinese styles—comprehensively capturing layout characteristics, furniture configurations, and functional usage scenarios across various room types.

Comprehensive Annotation Format. Leveraging algorithmic techniques and AIGC [51], we standardized the data pair organization to provide multi-dimensional, cross-modal supervision. We manually annotated approximately 20,000 scene samples, where each sample strictly follows a unified structure encompassing: a natural language instruction, structured CSS Scene Code, a scene relation graph capturing topological positional relationships, multi-view rendered images, and the corresponding 3D scene representation.

Rigorous Quality Control. To ensure high-fidelity alignment between textual semantics, local spatial parameters, and overall 3D rendering performance, we implemented a strict "human-machine collaborative validation" mechanism. Initially, automated scripts are utilized to globally detect and filter out physical violations, such as object interpenetration and dimensional anomalies. Subsequently, this process is complemented by manual random inspections to strictly guarantee the accuracy, consistency, and spatial rationality of the layouts, thereby establishing

a reliable foundation for downstream controllable generation tasks.

3.2.2 | Self-Correcting

Prompt engineering is a technique that guides the outputs of Large Language Models (LLMs) through the careful design of natural language instructions. In scene generation tasks, relying solely on initial text prompts often falls short in completely circumventing complex physical violations. To further enhance generation quality, inspired by works such as CC3D [52], this paper leverages the robust in-context learning and code comprehension capabilities of LLMs to innovatively design a self-correction mechanism based on prompt engineering. Specifically, rather than being confined to a single-pass forward generation, this work seamlessly embeds pre-designed quantitative evaluation metrics for "layout rationality" and "category rationality" into an iterative feedback loop. The system first performs geometric conflict validation on the initially generated scene layout, and subsequently encodes the computed anomalous features (e.g., object interpenetration ratios and boundary violations) into "rectification prompts," which are fed back into the LLM as additional inputs. Upon receiving this structured evaluative feedback, the LLM can autonomously execute local reorganization and spatial optimization. Through comprehensive ablation studies, this paper demonstrates that this self-correction mechanism, which integrates quantitative evaluation metrics with the interactive capabilities of language models, significantly enhances the physical feasibility and semantic fidelity of the final 3D scene generation.

3.2.3 | Evaluation Metrics

To fully leverage the LayoutVerse-20K dataset, we propose two key evaluation metrics that assess design schemes across two dimensions: spatial and categorical. The spatial metric P_L evaluates the geometric rationality of the layout, the relative positioning of objects, and spatial efficiency to ensure the feasibility and coherence of the scene in the physical environment. The categorical metric P_C assesses object classification accuracy, functional layout consistency, and alignment with user requirements, ensuring that the scene satisfies semantic-level functional demands. The technical implementation is divided into three stages:

Data preparation and feature extraction. A graph-based parsing algorithm is then used to extract the spatial layout matrix $G_l \in \mathbb{R}^{N \times N}$, where N denotes the number of spatial elements. Simultaneously, the material description matrix $T_c \in \mathbb{R}^{d \times m}$ is obtained, where d is the feature dimension and m is the number of object categories. Next, the reference set S_{ref} is constructed. Specifically, Sentence-BERT is employed to semantically align user prompts e_p with LayoutVerse-20K scene descriptions $\hat{e}_i$, which can be written as:

$$S_{ref} = \text{top } k \left(\arg \max_{\hat{e}_i \in E} \text{Similarity}(e_p, \hat{e}_i) \right). \quad (2)$$

Category plausibility assessment. BERT is used to encode the material description text $\hat{T}c$ for each scene in the reference set S_{ref} . The reference category plausibility matrix S_C is then constructed

as follows:

$$p_i = \text{softmax}(\text{Similarity}(e_p, \hat{e}_i)), S_c = \sum_{i=1}^k p_i * \hat{T}_c. \quad (3)$$

Finally, we calculate the similarity of S_c and T_c to obtain the category plausibility P_C as follows:

$$P_C = \text{Similarity}(T_c, S_c). \quad (4)$$

Layout plausibility assessment. We use the graph-structure parsing algorithm to extract the spatial layout matrix $\hat{G}_i$ of each scene in S_{ref} . The eigenvalues f_i of G_i are then extracted, and feature dimensions are unified through linear projection as follows:

$$\hat{f}_i = W * f_i + b \in \mathbb{R}^k. \quad (5)$$

Next, the reference layout vector S_l is obtained through weighted aggregation, and similarity computation is performed to derive the layout rationality score P_L as follows:

$$S_l = \sum_{i=1}^k p_i * \hat{G}_i, \quad (6)$$

$$P_L = \text{Similarity}(G_l, S_l). \quad (7)$$

3.3 | Scene Generation

In the previous section, the scene layout (CSS Scene Code) and a self-correction mechanism were constructed using Large Language Models and graph structure modeling, thereby establishing the physical and semantic "skeleton" of the scene. To further endow this skeleton with geometric textures [53] and continuous physical surfaces [54], this section addresses the bottlenecks of traditional single-stage incremental generation methods regarding computational overhead and the balance between texture and geometry [55, 56]. Consequently, based on prior research [57, 58, 59], we propose a two-stage decoupled generation paradigm: "2D visual prior first, followed by 3D Gaussian reconstruction."

This paradigm introduces the panorama as a "middleware" connecting the discrete layout with the continuous 3D space. This approach not only achieves seamless global modeling to eliminate the viewpoint and lighting inconsistencies [60] caused by multi-view stitching, but also effectively encapsulates and maps the preceding graph structural topology and boundary constraints. Specifically, the generation pipeline is divided into two stages: first, a diffusion model is utilized to generate a high-resolution panorama that is strictly aligned with the input instructions and layout, providing a visual prior; subsequently, by combining multi-view depth estimation [61] with point cloud fusion algorithms, the panorama is smoothly "lifted" into 3D and reconstructed into a high-fidelity 3D scene via 3D Gaussian Splatting technology [62]. The remainder of this section will elaborate on the core components, including panorama resolution enhancement, multi-view depth estimation, point cloud initialization and deduplication, and Gaussian rendering optimization.

Panorama Generation and Processing. Multi-Layer Perceptron (MLP) and Low-Rank Adaptation (LoRA) layers are first integrated into the PanFusion model to enhance its capability in processing complex and lengthy text descriptions, which can be written as:

$$h = \sigma(W_2 * \text{ReLU}(W_1 * e_T + b_1) + b_2) + U * \text{soft}(V * h), \quad (8)$$

where e_T denotes the corresponding text embedding, W_1 and W_2 are the weight matrices of the MLP layer, b_1 and b_2 are the bias terms, and σ is the activation function. U and V represent the low-rank matrices in the LoRA layer.

The low-resolution initial image initially lacks details [63], which will make the reconstructed point cloud sparse and blurry [64]. To improve the subsequent reconstruction quality [65], the scene graph is first super-resolution processed, and then the panorama is converted into a perspective view using the E2P algorithm. This process can be expressed as:

$$I_{\text{pers}}(u, v) = E(I_{\text{HR}}, f(u, v), R, T, S), \quad (9)$$

where $I_{\text{pers}}(u, v)$ denotes the pixel value of the perspective image at position (u, v) , and $f(u, v)$ is the mapping function from planar coordinates to spherical coordinates. The matrices R and T represent the camera's orientation and position, respectively. S denotes the size of the output image.

Point Cloud Construction with Gaussian Splatting. Given the multi-view images, the ZoeDepth model [66] is employed to estimate their corresponding depth maps D_i .

Subsequently, the point cloud is initialized using a cyclic generation strategy. Specifically, a function $f_{2 \rightarrow 3}$ is employed to derive the single-view point cloud position ρ_0 from the input image I_0 , depth map D_0 , and camera intrinsic matrix K :

$$\rho_0 = f_{2 \rightarrow 3}(I_0, D_0, K, P_0), \quad (10)$$

where P_0 represents the external matrix of I_0 . To eliminate duplicate points during the merging of point clouds from different viewpoints, a mask M_i is defined, and the optimal depth scaling factor d_i is estimated. Specifically, d_i is determined by minimizing the distance between the 3D points of the current image and their corresponding points in the previous point cloud ρ_{i-1} .

After removing duplicate points, the corrected depth map $d_i D_i$ is used to generate the new point cloud ρ_i as follows:

$$\rho_i = f_{2 \rightarrow 3}(I_i - M_i, d_i D_i, K, P_i). \quad (11)$$

Finally, the point clouds are iteratively fused to form the complete initial point cloud $\mathcal{R}$ as follows:

$$\mathcal{R} = \bigcup_{i=1}^{|P|} \phi(\rho_i, \rho_{i-1}), \quad (12)$$

where ϕ is the depth alignment function. After generating the initial point cloud, a model is trained using 3D Gaussian Splatting [67]. The initial point cloud, denoted as $\mathcal{R}$, serves as the set of initial Structure-from-Motion (SfM) points, which accelerates network convergence and enables the model to focus on generating detailed and realistic scene representations [68].

During the training process, multi-view image slices are used as input [69, 70, 71, 72], and the image view augmentation technique from LucidDreamer [73, 74, 75] is incorporated to enrich the dataset. Since the input multi-view images [76] may not fully cover the entire 3D space, the point cloud $\mathcal{R}$ is projected into 2D from multiple camera viewpoints. This effectively increases the

number of supervisory views and enhances spatial coverage. This approach leads to a more consistent and complete scene-level point cloud, enabling the generation of high-fidelity 3D scenes.

4 | Experiments

In this section, we conduct a comprehensive suite of experiments to validate the effectiveness and superiority of the proposed ControlScene. Our evaluations span both qualitative and quantitative analyses, aiming to assess the realism, semantic fidelity, and spatial coherence of the generated 3D scenes in response to natural language prompts. To ensure objective evaluation, we adopt three carefully designed metrics: Category Plausibility (P_C), Layout Rationality (P_L), and Semantic Similarity (α). Each of these metrics is aligned with a specific aspect of scene generation quality and grounded in prior work on text-to-3D synthesis. Our model could run on 1 NVIDIA 3090 with 24 GB memory.

Category Plausibility (P_C) measures the semantic appropriateness of the object categories within a generated scene. This metric evaluates whether the furniture or objects present in a layout are logically consistent with the type of room described in the prompt. For instance, placing a stove in a bathroom would yield a low P_C score, whereas correctly assigning beds to a bedroom improves the plausibility.

Layout Rationality (P_L) assesses the spatial configuration and geometric feasibility of the generated layout. This includes evaluating the relative positioning, alignment, and spacing of objects, as well as compliance with real-world spatial conventions. Higher values of P_L indicate layouts that are better aligned with architectural and ergonomic norms.

Semantic Similarity (α) quantifies how closely the generated scene matches the user's intent as expressed in the input text prompt. It is calculated using the Pearson correlation coefficient between the sentence embeddings of the input prompt ($\vec{e}_p$) and the generated scene description ($\vec{e}_r$). The α value is calculated as follows:

$$\alpha = r(\vec{e}_p, \vec{e}_r) = \frac{\sum_{i=1}^n (e_{pi} - \bar{e}_p)(e_{ri} - \bar{e}_r)}{\sqrt{\sum_{i=1}^n (e_{pi} - \bar{e}_p)^2} \sqrt{\sum_{i=1}^n (e_{ri} - \bar{e}_r)^2}}. \quad (13)$$

4.1 | Performance Comparison with LLM

We present a quantitative comparison between our proposed model, ControlScene, and both general-purpose large-scale language models (Qwen [77], Kolors [78], Doubao [79]) as well as domain-specific baselines (PanFusion [80], Text2Light [81]). The evaluation results, summarized in Table 2, include three key metrics: layout rationality (P_L), category plausibility (P_C), and semantic similarity (α). ControlScene achieves the highest scores across all three metrics, demonstrating clear advantages in spatial coherence, semantic alignment, and prompt fidelity. Specifically, ControlScene achieves a P_L score of 0.744, significantly outperforming all baselines and indicating its superior ability to generate geometrically coherent and well-structured layouts. For P_C , ControlScene scores 0.703, surpassing even the best-performing general models (Doubao and Kolors), and demonstrating strong

TABLE 2 | Performance Comparison of Different Models

Model	$P_L \uparrow$	$P_C \uparrow$	$\alpha \uparrow$
PanFusion [80]	0.621	0.588	0.576
Qwen2.5 [77]	0.639	0.653	0.695
Kolors [78]	0.699	0.675	0.681
Doubao [79]	0.534	0.697	0.697
Text2Light [81]	0.537	0.674	0.493
ControlScene(Ours)	0.744	0.703	0.744

semantic consistency in aligning scene content with common-sense object-role associations. Furthermore, in terms of semantic similarity ($\alpha = 0.744$), which measures the alignment between the generated scene and the input prompt, ControlScene again outperforms all competitors, affirming the effectiveness of our prompt tuning and refinement strategies.

4.2 | The Results of Self-Correction Mechanism

To intuitively verify the actual effect of the self-correction mechanism proposed in this paper in improving the quality of initial generation, this section specifically designs a comparative experiment based on process effectiveness analysis. This experiment focuses on the "intermediate state" of scene generation—namely, the structured scene sequence and spatial relation graph—aiming to strip away the visual interference of downstream 3D rendering and purely evaluate the large language model's self-rectification capability regarding spatial logic. In terms of experimental design, we selected 30 test samples covering various levels of reasonableness, strictly comparing the "original layout and graph structure" directly generated from the user's initial natural language instructions with the "final layout and graph structure" after iterative optimization via the self-correction loop. The experiment primarily quantifies the performance changes in two core dimensions, Category Reasonableness P_C and Layout Reasonableness P_L , to verify the mechanism's actual efficacy in eliminating physical conflicts and functional fallacies.

As shown in the left sub-figure of Fig. 3, this experiment deeply quantifies the framework's self-rectification performance at the "intermediate state" level by strictly comparing the initial layout and graph structure automatically parsed after user input with the final versions optimized through loop iteration. The results indicate that after the introduction of the self-correction mechanism, both key metrics show a highly significant upward trend across the entire dataset:

Regarding Category Reasonableness P_C , which reflects the logic of the graph structure, the average score of the samples leaped from 0.493 in the initial generation to 0.744. This data suggests that when semantic inconsistencies or nodes violating common sense appear in the initially extracted scene graph (for example, a "refrigerator" node being abnormally associated with the graph structure of a "bedroom"), the self-correction loop can effectively identify and replace these logically disordered objects using the large model's knowledge base, ensuring that the layout remains functionally consistent with the intended room context.

Regarding Layout Reasonableness P_L , which reflects the geometric dimension of the scene sequence, the self-correction mechanism demonstrates a more breakthrough-style spatial restructuring capability, with the average score significantly improving

from 0.450 to 0.703. For low-quality samples characterized by cluttered bounding box stacks or severe interpenetration and floating of parametric coordinates, the mechanism extensively restructures spatial parameters based on quantitative feedback from physical collisions, mathematically correcting object spacing, orientation, and alignment with architectural boundaries. For samples with relatively refined initial spatial parameters, the self-correction mechanism still provides delicate local coordinate optimizations, such as enhancing symmetry and aligning the visual focal points of furniture with the room, thereby comprehensively improving overall spatial aesthetics and realism at the code level.

Fig. 4 intuitively illustrates the actual performance of the self-correction mechanism in handling typical spatial conflicts and logical fallacies. As shown in the top-left case of Figure 4, in the initially generated bedroom scene, the desk and chair nodes exhibited obvious physical overlap (interpenetration), and the desk was floating in violation of common sense; after the intervention of the self-correction mechanism, the model precisely repaired the desk's spatial coordinates and grounding relationship by analyzing physical collision feedback. As shown in the bottom-left case, addressing the severe semantic conflict where a "desk" erroneously appeared in a shower room, the self-correction mechanism identified and removed the conflicting furniture through common-sense auditing of the scene graph. These results, based on intermediate structured data, fully confirm that the self-correction mechanism does not rely on backend visual rendering retouching but instead operates at the level of the underlying physics engine and logical semantics. It not only accurately "saves" poor designs with severe conflicts but also "refines" already reasonable geometric topologies, serving as a robust enhancement module that effectively eliminates LLM "spatial hallucinations" and provides a solid "foundation" for high-fidelity downstream 3D rendering.

This paper presents a comparison of panoramic visual instantiation between the baseline variant and the full system under the same prompt in Figure 5.

In the baseline model pipeline lacking the self-correction mechanism, as shown in the upper part of Figure 5, the system is unable to perform physical and common-sense auditing on the initial layout output by the large language model. This leads to the erroneous generation and rendering of a "desk"—a piece of furniture that severely violates daily common sense and functional logic—within the specific functional space of a "bathroom" (indicated by the red-boxed area in the figure). Such underlying semantic conflicts not only undermine the rationality of the scene but also irreversibly pollute the downstream panoramic image synthesis and 3D reconstruction results.

In contrast, after the introduction of the closed-loop self-correction mechanism, as shown in the lower part of Figure 5, the system successfully identifies this logical fallacy during the underlying structured data stage and automatically eliminates the conflicting node using the prior knowledge base. Comparing the red-boxed area at the bottom of the figure, it is clearly visible that the originally abrupt desk has been reasonably removed, and the overall spatial topology and visual semantics of the bathroom have been perfectly maintained. This intuitive comparison fully demonstrates that the self-correction mechanism is not a

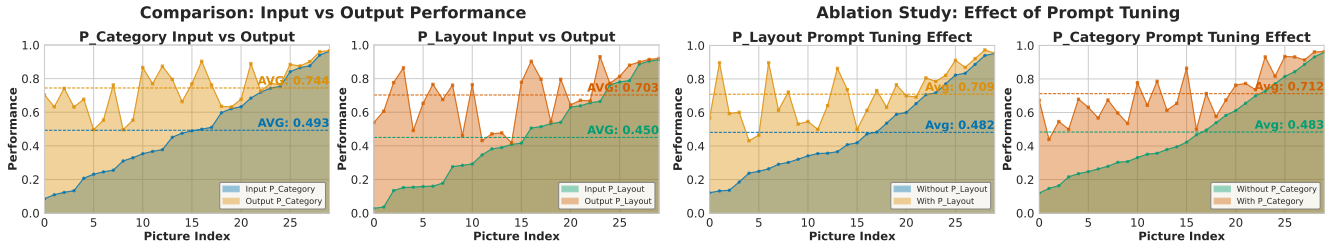


FIGURE 3 | Quantitative evaluation of our method. The two plots on the left compare the input and output performance in terms of category plausibility (P_C) and layout rationality (P_L), showing significant improvements after applying our self-correction mechanism. The two plots on the right present ablation studies analyzing the effect of prompt tuning.

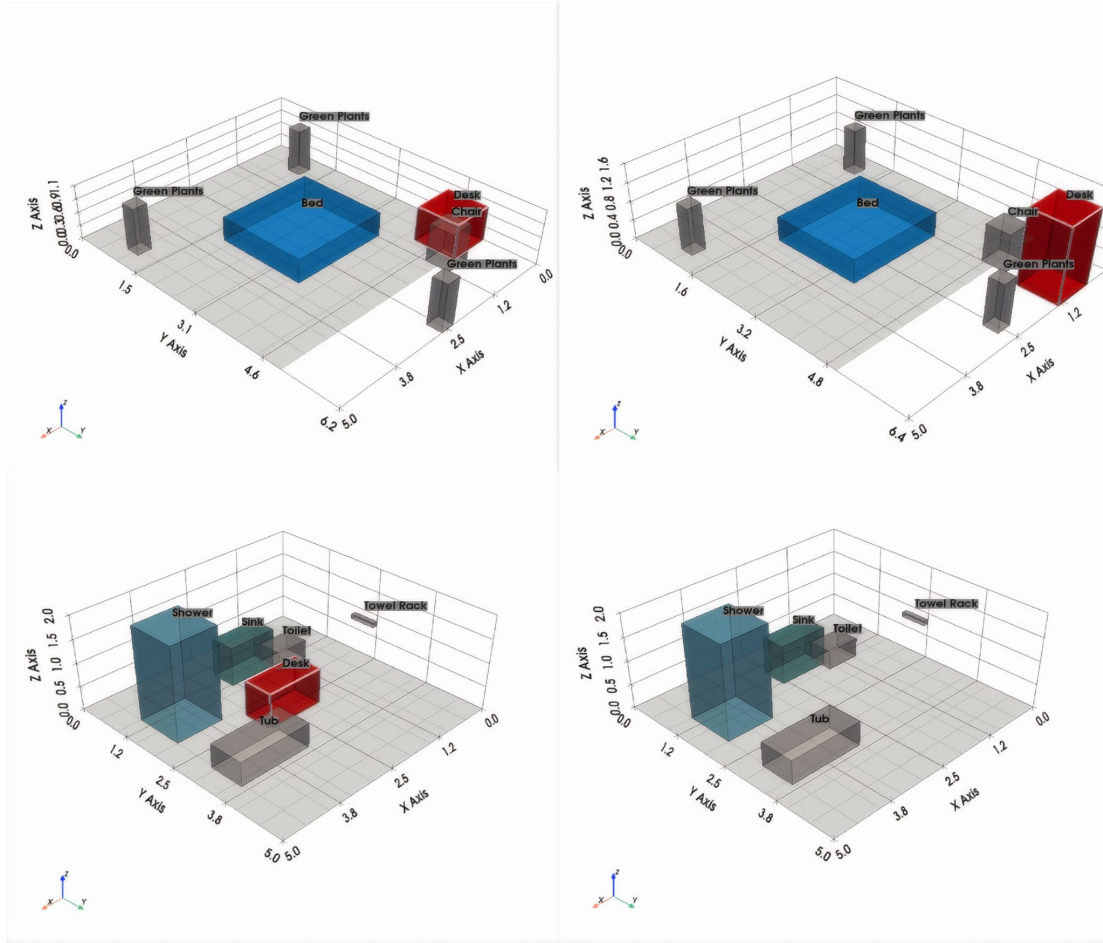


FIGURE 4 | Showcase of self-correction mechanism cases.

post-processing modification, but a core hub that curbs generation deviation at the source and guarantees the functional logic consistency of multi-modal scenes.

4.3 | Ablation Study

The effectiveness of LLM. We conducted an ablation experiment comparing the performance of our framework with and without the LLM component, denoted as “W/ LLM” and “W/o LLM”, respectively. Both variants were evaluated under identical input conditions to ensure experimental consistency. As shown

in Table 3, the version augmented with the LLM achieves significantly higher performance across all test samples, with an average score of 0.744 compared to 0.457 for the non-LLM baseline. This substantial margin highlights the LLM’s critical role in enhancing both the semantic alignment and contextual relevance of generated scenes. Specifically, the LLM effectively interprets complex natural language prompts and contributes to more accurate object placement, spatial arrangement, and category selection. The increased similarity between user intent and final output strongly suggests that the LLM enhances the expressiveness, coherence, and fidelity of scene generation. These results



FIGURE 5 | Comparison of visual instantiation between the baseline variant (top) and the self-correction mechanism (bottom)

underscore the importance of embedding LLM into the generation loop to better capture linguistic subtleties and translate them into semantically rich, spatially coherent 3D environments. The effectiveness of self-correction mechanism. To rigorously evaluate the effectiveness of our proposed self-correction mechanism, we conducted an ablation study by comparing the full model against a variant with the self-correction component removed, while keeping all input conditions identical. As shown in Fig. 3, the model incorporating the self-correction mechanism consistently outperforms the baseline across all evaluated metrics. In terms of layout plausibility, the corrected model demonstrates a clear advantage, with an average performance of 0.709 compared to 0.482, indicating its ability to generate spatially coherent layouts with improved object alignment, spacing, and architectural adherence. Similarly, for category plausibility, the average performance increases from 0.483 to 0.712 when self-correction is enabled, suggesting enhanced semantic alignment of objects with their functional roles and contexts. These improvements are particularly notable in scenes with initially ambiguous or low-quality structures, where the mechanism effectively restructures and refines the spatial and categorical composition. Even in already reasonable scenes, it contributes subtle yet meaningful enhancements that bring outputs closer to real-world expectations. Overall, these findings underscore the self-correction

module's crucial role in boosting the fidelity and practical reliability of the ControlScene framework.

4.4 | Visualization Analysis

The results in Fig. 6 illustrate the strong visual consistency and semantic fidelity of our generation pipeline. Given complex and descriptive prompts, our model is able to accurately instantiate both prominent and supporting scene elements in the panoramic image, while maintaining logical spatial configurations across multiple viewpoints through 3D Gaussian Splatting. For example, the blue quilt and small lamp in the bedroom scene appear exactly where expected relative to the bed, confirming adherence to common sense and user intent. Similarly, the sectional sofa and TV are correctly positioned in the family room, and specific details such as cow-themed stuffed toys and a blue cushioned bench are faithfully generated in the nursery. These results verify that ControlScene not only interprets linguistic cues accurately but also preserves object relationships across 3D reconstructions, enabling realistic, explorable scenes directly from text.

TABLE 3 | Comparison of α with and without LLM

Index	W/o LLM	W/ LLM	Index	W/o LLM	W/ LLM
1	0.014	0.631	11	0.290	0.637
2	0.042	0.672	12	0.318	0.661
3	0.058	0.677	13	0.362	0.686
4	0.086	0.612	14	0.390	0.634
5	0.130	0.576	15	0.406	0.599
6	0.158	0.733	16	0.434	0.669
7	0.174	0.633	17	0.478	0.776
8	0.202	0.634	18	0.506	0.669
9	0.246	0.863	19	0.550	0.732
10	0.274	0.716	20	0.594	0.764

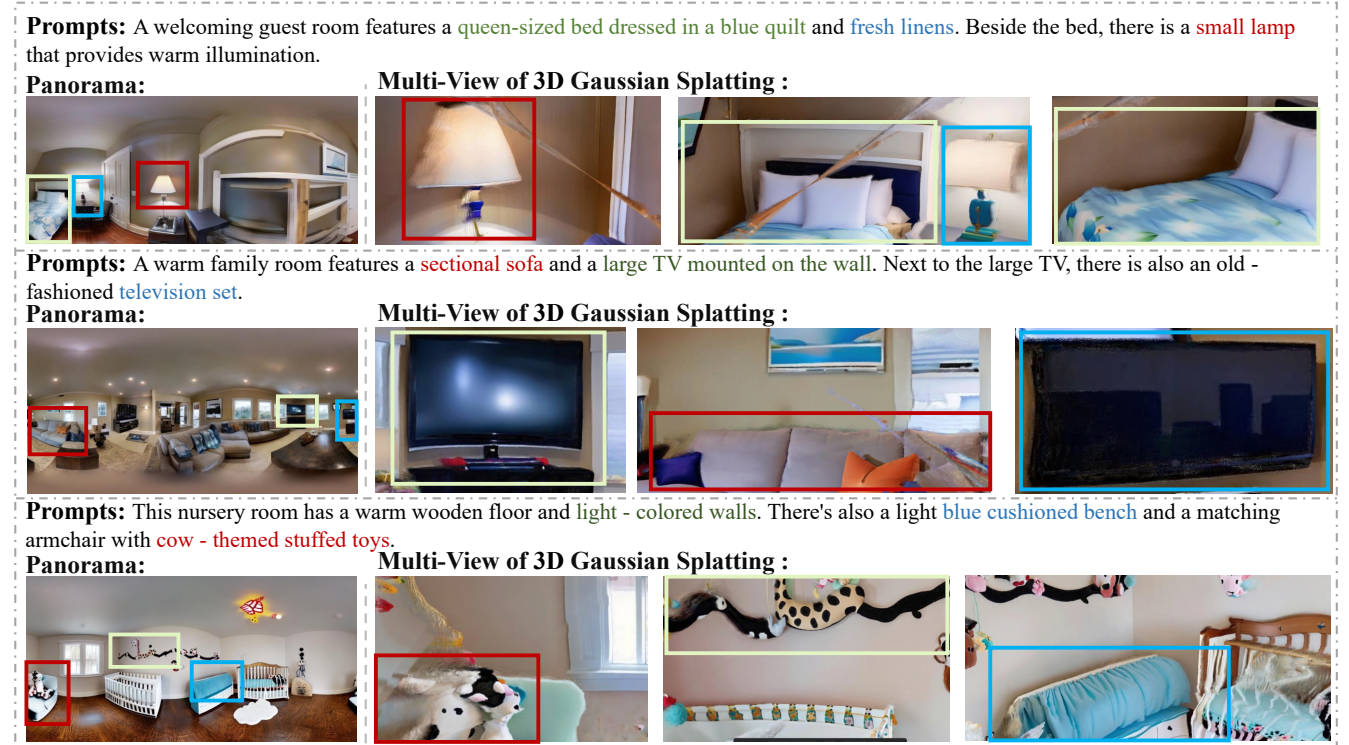


FIGURE 6 | Qualitative visualization results: Each row shows a text prompt and its corresponding panorama and 3D Gaussian Splatting rendered multi-view image. The key objects in the prompt are accurately synthesized and highlighted in both the 2D panorama and 3D reconstructed views.

5 | Discussion and Limitations

While ControlScene demonstrates robust capabilities in semantic faithfulness and spatial coherence, there are still several limitations and avenues for future exploration.

Balancing Statistical Priors and Creative Generation. To prevent severe physical hallucinations—such as the "Penrose Triangle Problem" (e.g., generating a desk inside a shower room)—our framework relies on a statistical prior derived from Top-K retrieval. While this mechanism serves as a robust foundational defense to ensure baseline spatial rationality, we acknowledge that an over-reliance on statistical priors may inadvertently penalize highly creative or long-tail valid prompts. To mitigate this constraint on creativity, the current system integrates an interactive graphical user interface (GUI) that allows users to flexibly intervene and override parameters when the statistical prior restricts their specific creative intent. In future work, we plan to introduce a dynamic weight evaluation module driven by the

commonsense reasoning capabilities of Large Language Models, which will better balance statistical plausibility with generative creativity.

System Usability and Subjective Evaluation. The core objective of ControlScene is to propose an algorithmic framework that addresses spatial hallucinations and physical conflicts in 3D scene generation through a self-correction mechanism and structured layout priors. The accompanying interactive UI primarily serves as an intuitive validation platform to demonstrate the fine-grained editability of the CSS Scene Code and the system's potential for human-AI co-creation. To objectively evaluate the system's effectiveness without subjective bias, we designed two fine-grained quantitative metrics: category plausibility (P_C) and layout plausibility (P_L). Extensive ablation studies firmly validate the system's superior capability in responding to local modifications and resolving spatial conflicts. Nevertheless, we acknowledge that a comprehensive evaluation of system usability and

human-computer collaboration experience requires a rigorous user study (e.g., measuring task completion time and System Usability Scale scores). Since this entails a broad user base and complex comparative tasks more aligned with dedicated Human-Computer Interaction (HCI) research, we reserve this large-scale subjective evaluation for future work.

Spatial Error Accumulation in 2D-to-3D Lifting. During the transition from the discrete CSS Scene Code to the continuous 3D physical space, there remains a possibility of slight error accumulation. To maximally constrain this drift, our system enforces strict design mechanisms: the generated high-resolution panoramas are rigidly conditioned on the pre-generated layouts, acting as strong visual priors. Furthermore, during the ZoeDepth-based multi-view point cloud construction phase (encompassing PanFusion, super-resolution, E2P projection, and point cloud de-duplication), a multi-view geometric consistency algorithm ensures that relative positions, scale proportions, and occlusion relationships are preserved without fundamental semantic shifts. Although minor pixel-level deformations might still occur at the extreme edges of depth boundaries, the overall topological structure and the physical relationships defined by the user are strictly maintained. Addressing these boundary deformations to achieve pixel-perfect spatial fidelity will be a focus of our subsequent optimization efforts.

6 | Conclusions

In this study, we propose ControlScene, an innovative framework focused on controllable text-to-3D scene generation. By integrating layout-level structures, ControlScene enables large language models to reason about both semantics and spatial layouts simultaneously, thus generating 3D scenes that are semantically faithful, spatially coherent, and user-controllable. Additionally, we release LayoutVerse-20K, a large-scale benchmark dataset containing 20,000 manually annotated samples. Each sample covers text prompts, layout metadata, graph-structured layouts, scene graphs, and corresponding 3D scenes. ControlScene demonstrates significant advantages over baseline methods in terms of spatial realism, semantic consistency, and user controllability, laying a solid foundation for future research in the field of language-driven 3D scene generation. Meanwhile, the LayoutVerse-20K dataset we proposed also provides more possibilities for research in related fields.

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